OXFORD
INTERNATIONAL
AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________
I declare this is my own work.

INTERNATIONAL GCSE
PHYSICS
Paper 1

Tuesday 8 November 2022 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a pencil and a ruler
- a scientific calculator
- the Physics Equations Sheet (enclosed).

Instructions
- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information
- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

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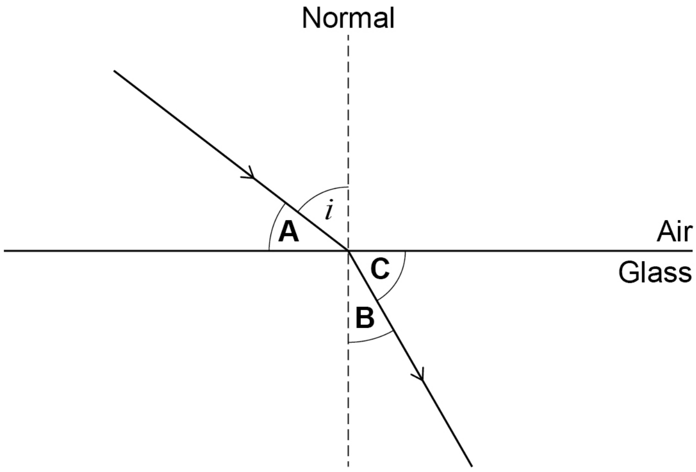
Answer **all** questions in the spaces provided.

0 1 A student investigated refraction.

Figure 1 shows light refracting as it passes from air into glass.

Figure 1



0 1 . 1 The angle of incidence is labelled i.

Which letter in **Figure 1** represents the angle of refraction?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

Do not write outside the box

0 1 . 2 The student measured the angle of incidence, i, and the angle of refraction, r.

The student then calculated the sine of each angle.

$\sin i = 0.70$

$\sin r = 0.50$

Calculate the refractive index of the glass.

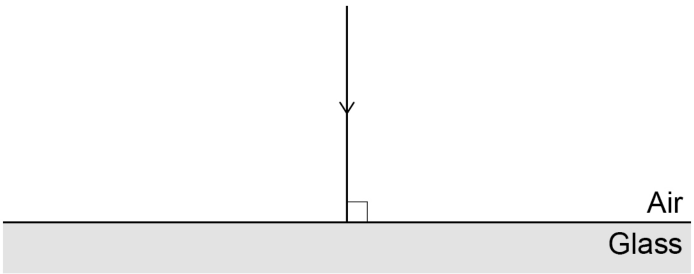
Use the Physics Equations Sheet.

[2 marks]

Refractive index = ____________

0 1 . 3 **Figure 2** shows a ray of light at 90° to the surface of the glass.

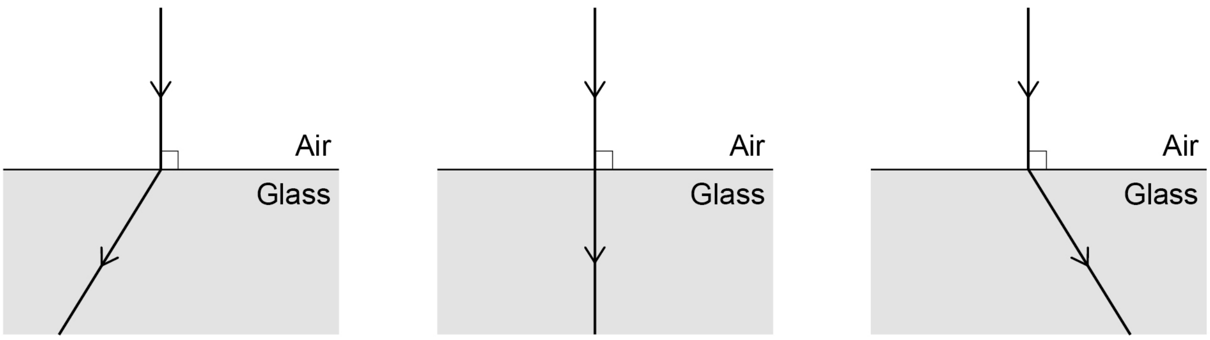
Figure 2



Which of the following shows what happens to the ray of light in **Figure 2** when it goes into the glass?

[1 mark]

Tick (✓) **one** box.



☐ ☐ ☐

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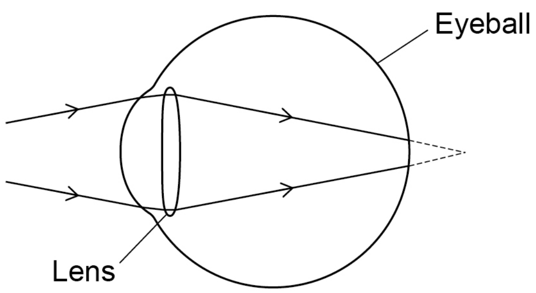
Light is refracted by the lens in an eye.

Figure 3 shows an eye with a vision defect.

Figure 3



0 1 . 4 What is the function of the retina?

[1 mark]

Tick (✓) **one** box.

To change the shape of the lens. ☐

To control the amount of light entering the eye. ☐

To detect light at the back of the eye. ☐

To refract light as it enters the eye. ☐

0 1 . 5 What causes the vision defect shown in **Figure 3**?

[1 mark]

Tick (✓) **one** box.

Cornea is too curved ☐

Eyeball is too long ☐

Eyeball is too short ☐

Lens is too powerful ☐

Do not write outside the box

0 1 . 6 A pair of glasses can be used to correct the vision defect shown in **Figure 3**.

What type of lens is used in these glasses?

[1 mark]

Tick (✓) **one** box.

Concave ☐

Convex ☐

Diverging ☐

0 1 . 7 Some eye defects can be corrected by laser eye treatment.

Complete the sentence to describe a laser.

Choose the answer from the box.

[1 mark]

light	**microwaves**	**sound**	**x-rays**

A laser is a source of ______________________.

0 1 . 8 Which of the following is another medical use of lasers?

[1 mark]

Tick (✓) **one** box.

To cauterise blood vessels. ☐

To create an image of an unborn baby. ☐

To detect broken bones. ☐

9

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

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0 2 A student investigated the melting point of coconut oil.

The student put some solid coconut oil into a test tube and melted it.

0 2 . 1 Describe how the student could melt the coconut oil safely.

[2 marks]

The student then measured the temperature of the coconut oil every 20 seconds as it cooled.

0 2 . 2 Give **two** pieces of measuring equipment that were needed by the student.

[2 marks]

1

2

Question 2 continues on the next page

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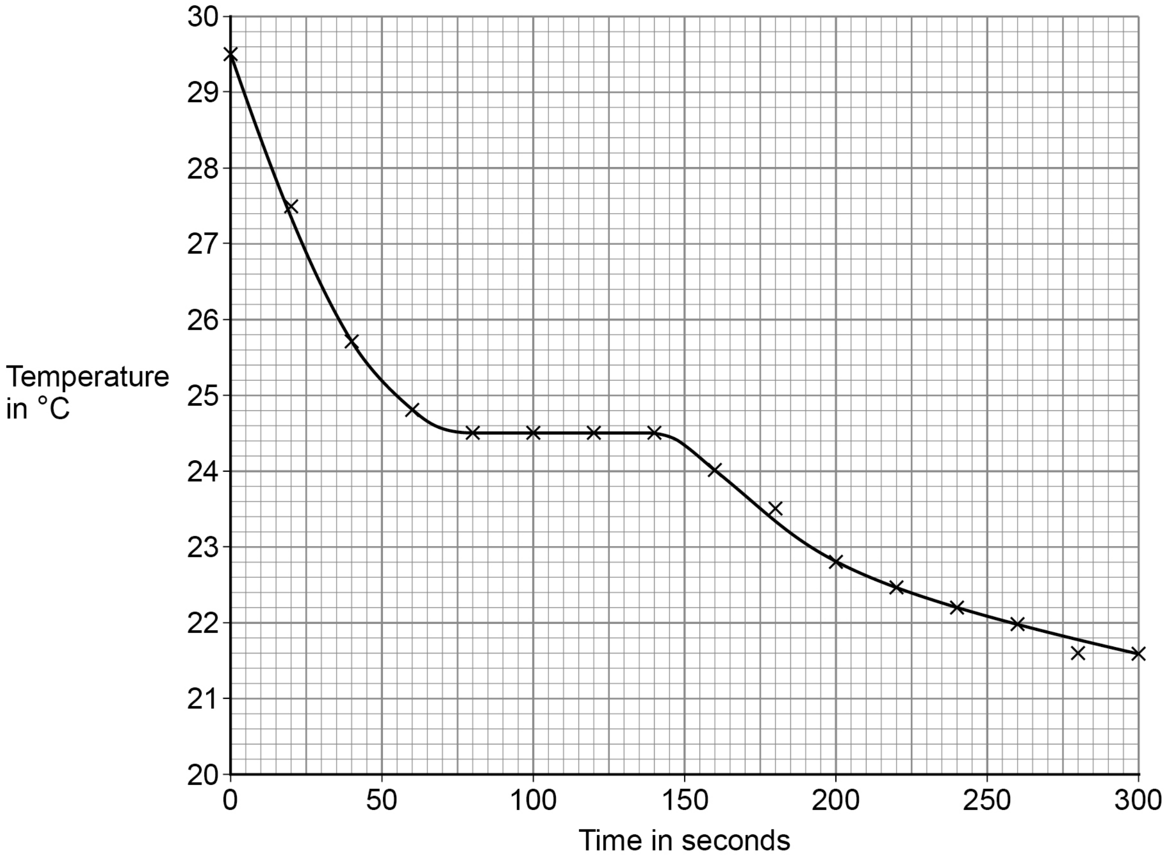
Figure 4 shows the results.

Figure 4



0 2 . 3 For how many seconds was the temperature of the coconut oil constant?

Use **Figure 4**.

[1 mark]

Tick (✓) **one** box.

25 s ☐

60 s ☐

120 s ☐

140 s ☐

Do not write outside the box

0 2 . 4 What is the melting point of the coconut oil?

Use **Figure 4**.

[1 mark]

Melting point = _______________ °C

0 2 . 5 Determine the change in temperature from the start of the investigation to when the coconut oil became solid.

Use **Figure 4**.

[1 mark]

____

____

Change in temperature = _______________ °C

0 2 . 6 The mass of the coconut oil was 0.080 kg.

specific latent heat of fusion of coconut oil = 110 000 J/kg

Calculate the energy transferred from the coconut oil as it changed from liquid to solid.

Use the Physics Equations Sheet.

[2 marks]

____

____

____

Energy transferred = _______________ J

0 2 . 7 Explain why the graph in **Figure 4** is a curve after 150 s.

[2 marks]

____

____

____

____

11

Turn over for the next question

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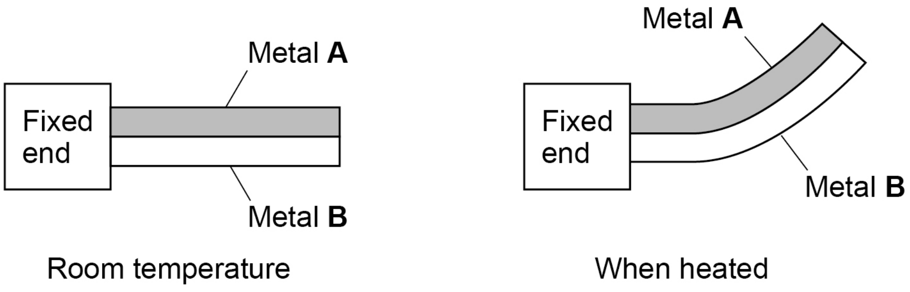
ANSWER IN THE SPACES PROVIDED

0 3 Metals expand when heated.

A bimetallic strip is made from two lengths of different metals which are joined together.

Figure 5 shows a bimetallic strip when it is at room temperature and when it has been heated.

Figure 5



0 3 . 1 Why does the bimetallic strip bend when it is heated?

[1 mark]

___

___

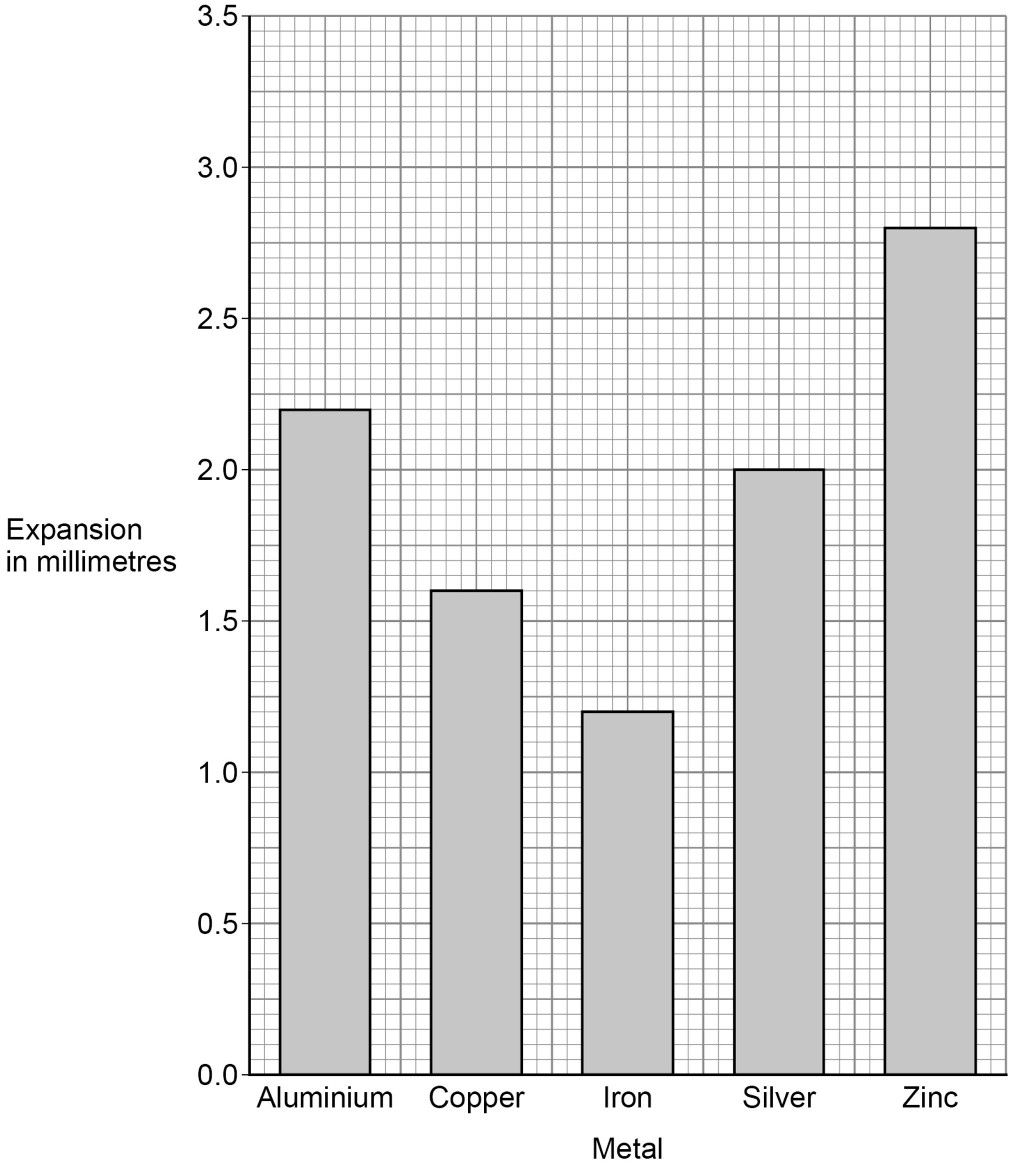
Question 3 continues on the next page

Five strips of different metals were heated for the same time.

The strips of metal were all the same length.

Figure 6 shows a bar chart of the five metal strips and expansion.

Figure 6



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0 3 . 2 Which **two** metal strips in **Figure 6** would bend the most when used as a bimetallic strip?

Give a reason for your answer.

[2 marks]

Metal strip 1 ______________________ Metal strip 2 ______________________

Reason ____

____

____

0 3 . 3 The original length of the aluminium strip is 700 mm.

Determine the percentage increase in length of the aluminium strip.

Use **Figure 6**.

[2 marks]

____

____

____

Percentage increase = ______________ %

Question 3 continues on the next page

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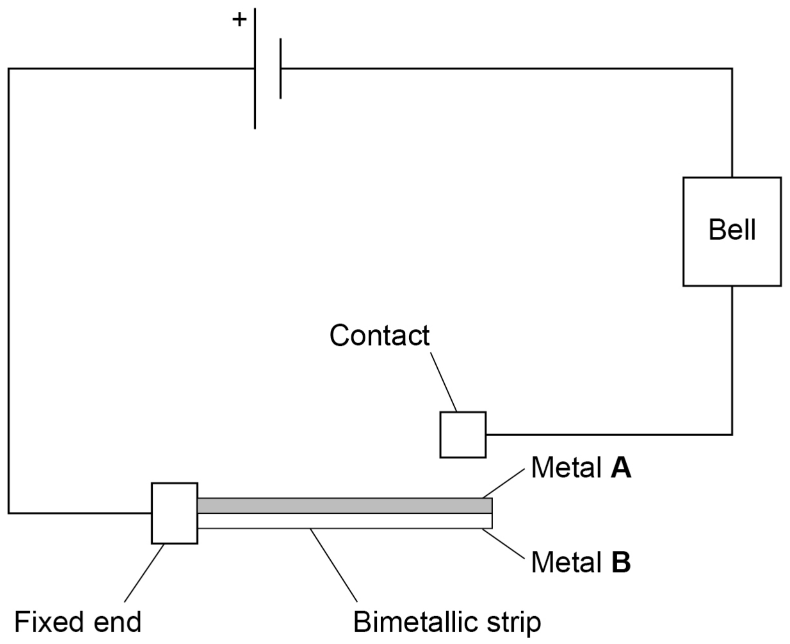
Figure 7 shows a bimetallic strip used in a fire alarm circuit.

Figure 7



0 3 . 4 Explain how the bimetallic strip switches the fire alarm on when there is a fire.

[2 marks]

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0 3 . 5 Explain why metals expand when heated.

[2 marks]

____

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9

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0 4 **Figure 8** shows a sign that displays the speed limit and a device that measures the speed of a passing car.

Figure 8

The device transmits microwaves.

The microwaves have a frequency of 24 GHz.

0 4 . 1 What is meant by 'frequency'?

[1 mark]

____

____

Question 4 continues on the next page

Turn over ►

0 4 . 2 Which of the following is the same as 24 GHz?

[1 mark]

Tick (✓) **one** box.

24 000 Hz ☐

24 000 000 Hz ☐

24 000 000 000 Hz ☐

24 000 000 000 000 Hz ☐

0 4 . 3 Calculate the wavelength of the microwaves emitted by the device.

speed of microwaves = 300 000 000 m/s

Use your answer from Question **04.2** and the Physics Equations Sheet.

[3 marks]

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____

Wavelength = __________________ m

The speed of the car is shown on the LED display.

Light waves are emitted from the LED display.

Light waves and microwaves are both electromagnetic waves.

0 4 . 4 Give **two other** similarities between light waves and microwaves.

[2 marks]

1 ____

____

2 ____

____

0 4 . 5 Give **one** difference between the properties of light waves and the properties of microwaves.

[1 mark]

____

____

Question 4 continues on the next page

Do not write outside the box

The sign in **Figure 8** on page 15 is used to encourage people to drive at a safe speed.

0 4 . 6 Speed affects the braking distance of a car.

Give **two other** factors that affect the braking distance of a car.

[2 marks]

1 ______________________________

____

2 ______________________________

____

0 4 . 7 Explain how driving at a slower speed decreases the **stopping** distance of a car.

[3 marks]

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13

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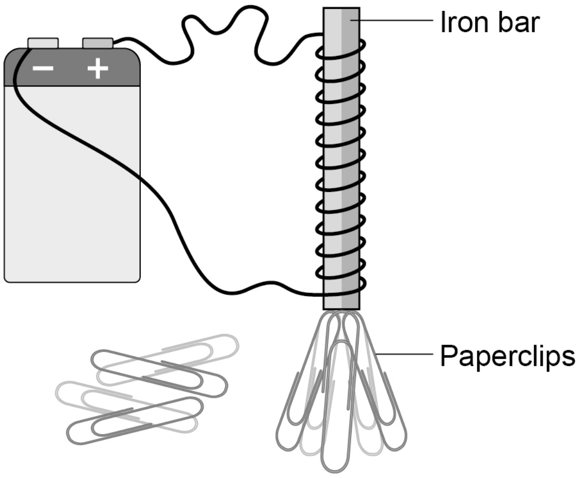
0 5 **Figure 9** shows a simple electromagnet being used to pick up some paperclips.

Figure 9



0 5 . 1 Give **one** advantage of electromagnets compared with permanent magnets.

[1 mark]

____

____

0 5 . 2 Give **two** changes that would cause the electromagnet in **Figure 9** to pick up more of the paperclips.

[2 marks]

1 ____

____

2 ____

____

Question 5 continues on the next page

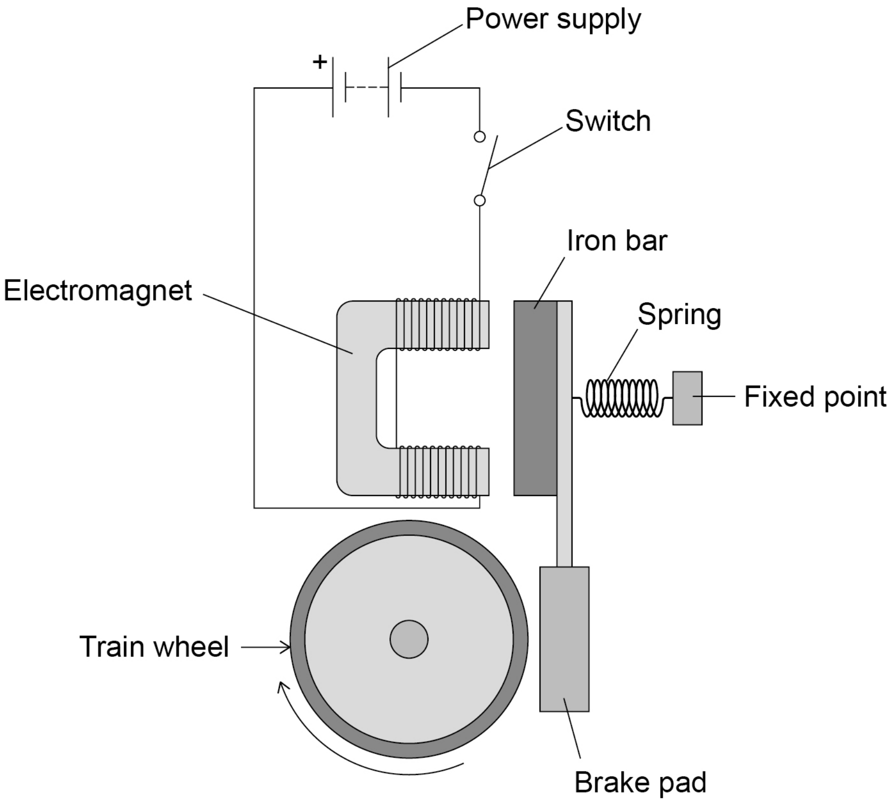
Do not write outside the box

0 5 . 3 An electromagnet can be used in an emergency braking system on a train.

In an emergency a switch is closed and then the moving train will come to a stop.

Figure 10 shows a design for the braking system on a train.

Figure 10



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Explain why the train will stop when the switch is closed.

[6 marks]

9

Turn over for the next question

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0 6 The Sun is a main sequence star.

0 6 . 1 Describe the process that happens in the core of the Sun and releases energy.

[3 marks]

0 6 . 2 Why does this process need very high temperatures?

[1 mark]

Do not write outside the box

0 6 . 3 Describe what will happen to the Sun after it reaches the end of the main sequence until the end of its life cycle.

[4 marks]

0 6 . 4 Stars behave like objects that emit black-body radiation.

What is meant by 'black-body radiation'?

[1 mark]

Question 6 continues on the next page

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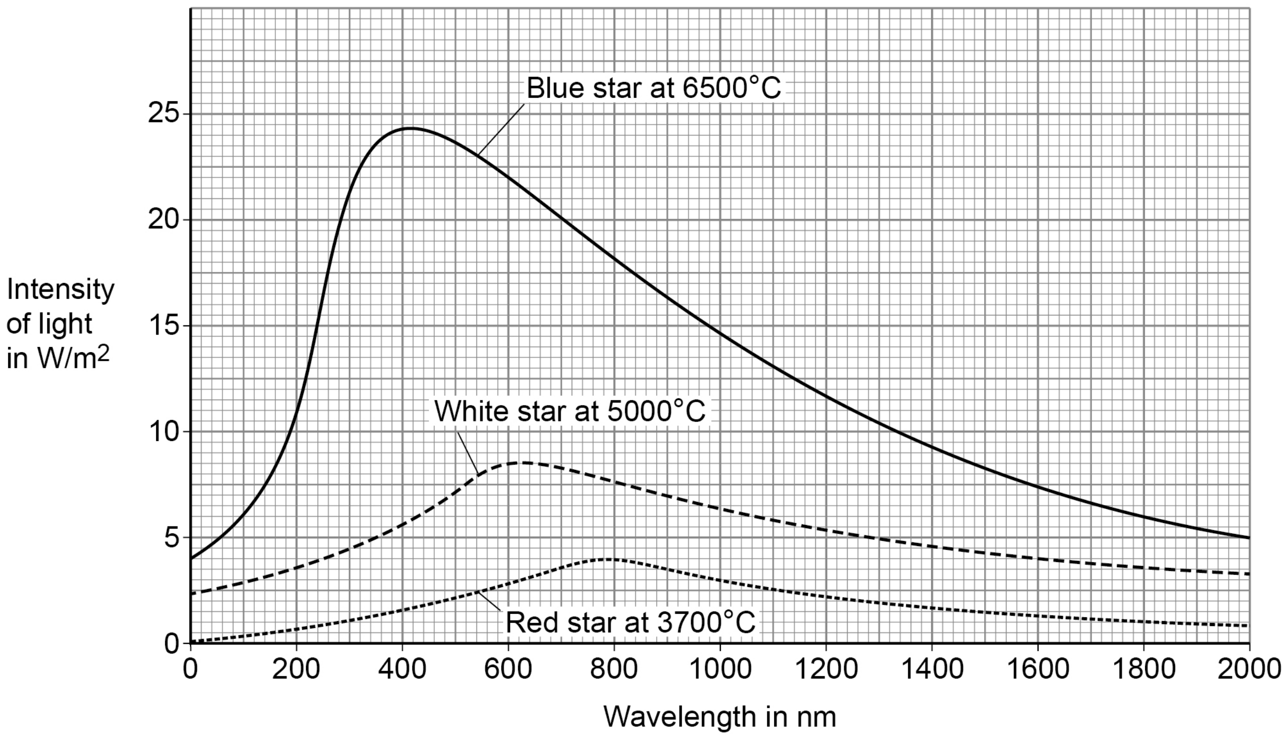
0 6 . 5 Stars are different colours because they have different surface temperatures.

Figure 11 shows how the intensity of electromagnetic radiation emitted varies with wavelength for three stars.

Figure 11



Compare how the intensity varies with wavelength for the three stars.

[4 marks]

13

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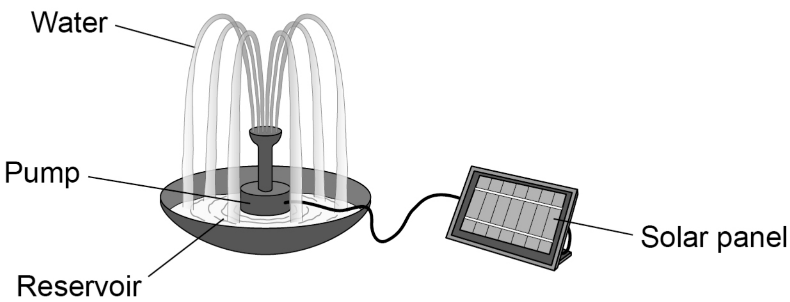
0 7 **Figure 12** shows a solar-powered fountain.

Water is pumped from the reservoir, sprays out from the fountain and returns to the reservoir.

Figure 12



0 7 . 1 Give **one** advantage of the fountain being solar-powered rather than connected to mains electricity.

[1 mark]

____

____

0 7 . 2 The solar panel has an efficiency of 0.30

Calculate the total energy in to the solar panel when the useful energy out is 12 000 J.

Use the Physics Equations Sheet.

[3 marks]

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Total energy = ____________________ J

Question 7 continues on the next page

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0 7 . 3 The solar panel also charges a battery during the day so the fountain will work at night.

The solar panel transfers 3000 C of charge to the battery.

potential difference across battery = 6.0 V

Calculate the energy transferred to the battery.

Use the Physics Equations Sheet.

[3 marks]

Energy transferred to battery = ____________________ J

Do not write outside the box

0 7 . 4 A drop of water of mass 2.0×10^{-4} kg leaves the fountain at a speed of 3.5 m/s.

Calculate the maximum height the drop of water could reach.

gravitational field strength = 9.8 N/kg

Use the Physics Equations Sheet.

[5 marks]

Maximum height = ____________________ m

0 7 . 5 The drop of water does not reach the height calculated in Question **07.4** because energy is dissipated to the surroundings.

Why is energy dissipated to the surroundings?

[1 mark]

13

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0 8 A drag race is a car race in a straight line, for a short distance.

Figure 13 shows a drag car used in a drag race. The car has parachutes at the back which open at the end of the race when the brakes are applied.

Figure 13

Question 8 continues on the next page

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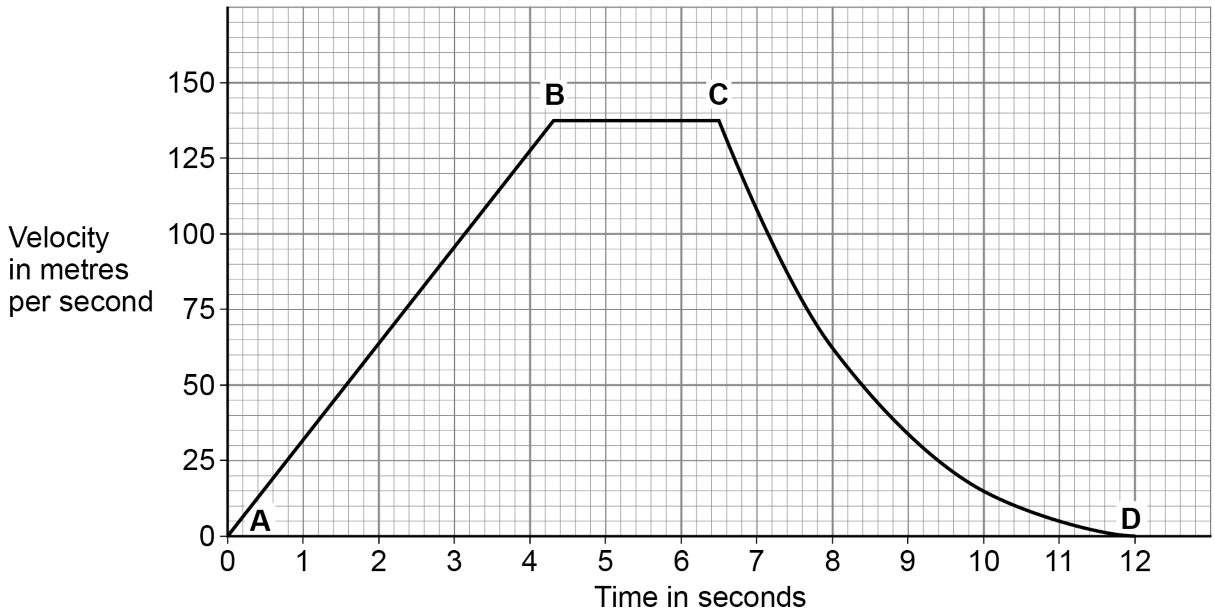
Figure 14 shows a velocity-time graph for the drag car during the race.

Figure 14



0 8 . 1 Explain the shape of the different sections of the graph in **Figure 14**.

Use ideas about forces in your answer.

[6 marks]

Section **A** to **B** ______________________________

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____

Section **B** to **C** ______________________________

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Section **C** to **D** ______________________________

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0 8 . 2 Determine the acceleration of the drag car during the first 1.4 s of the race.

Give the unit.

[3 marks]

Acceleration = ______________ Unit ____________

0 8 . 3 Determine the distance travelled by the drag car until the parachutes are opened.

[4 marks]

Distance travelled = ______________ m

13

END OF QUESTIONS

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